

Taylor Series for Approximating Nonlinear Devices

A Taylor series is a **series expansion** of a **function** about a point. A one-dimensional Taylor series is an expansion of a **real function** $f(x)$ about a point $x = a$ is given by

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \dots \quad (1)$$

If $a = 0$, the expansion is known as a **Maclaurin series**.

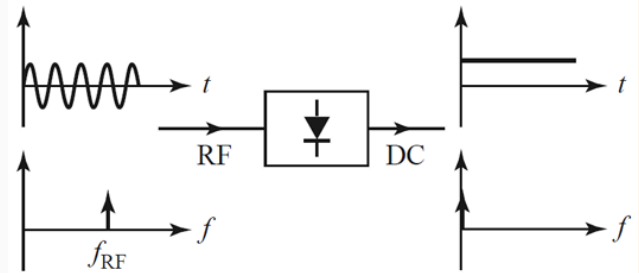
- $x - a$ = small signal vac, a is the biasing point (can be used for forward biased diode)
- for diode: $I(V) = I_s(e^{aV} - 1)$

Diode for Rectification

$$v = A \cos \omega_{RF} t$$

- The first two terms of the non-linear characteristic will give

$$\begin{aligned} i &= a_1 A \cos(\omega_{RF} t) + a_2 A^2 \cos^2(\omega_{RF} t) \\ &= a_1 A \cos(\omega_{RF} t) + \frac{1}{2} a_2 A^2 [1 + \cos(2\omega_{RF} t)] \\ &= a_1 A \cos(\omega_{RF} t) + \frac{1}{2} a_2 A^2 + \frac{1}{2} a_2 A^2 \cos(2\omega_{RF} t) \end{aligned}$$



➤ The DC rectified current: $i_{DC} = \frac{1}{2} a_2 A^2$

Diode for AM Demodulation

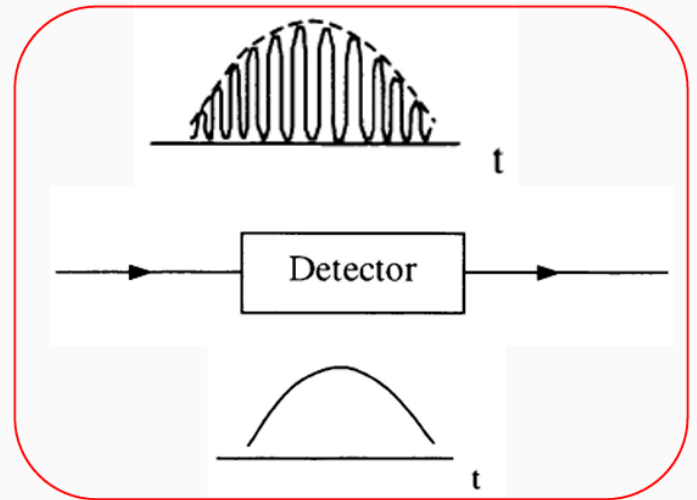
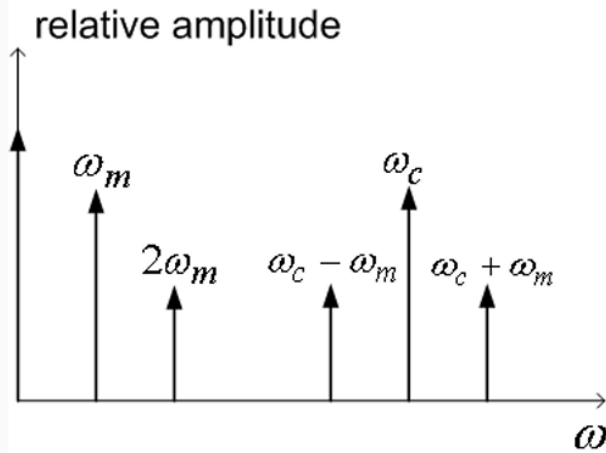
An amplitude modulated (AM) continuous wave is incident to the detector diode, we have

$$v = A(1 + m \sin \omega_m t) \sin \omega_c t$$

We assume the detector operates at the “**Square-law**” region. $i = a_1 v + a_2 v^2$

The desired output after the LPF

$$i_{out} = a_2 A^2 m \sin \omega_m t$$



Diode for Mixer

$$v_{RF}(t) = V_{RF} \cos \omega_{RF} t \text{ and } v_{LO}(t) = V_{LO} \cos \omega_{LO} t$$

$$i(t) = I_0 + a_1 [v_{RF}(t) + v_{LO}(t)] + a_2 [v_{RF}(t) + v_{LO}(t)]^2 + \dots$$

➤ $a_1 [v_{RF}(t) + v_{LO}(t)]$ is a replication of the RF and LO input signals, which will be filtered out by the low-pass IF filter.

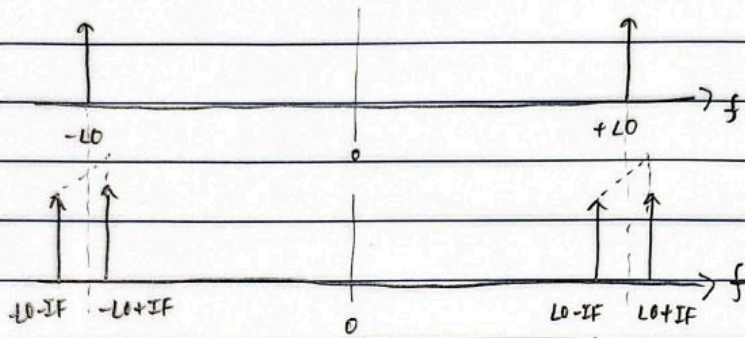
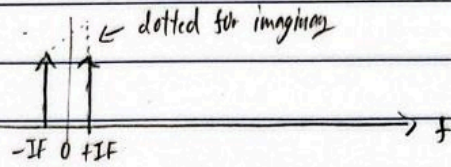
➤ The third term:

$$\begin{aligned} i(t) &= a_2 [V_{RF} \cos \omega_{RF} t + V_{LO} \cos \omega_{LO} t]^2 \\ &= a_2 [V_{RF}^2 \cos^2 \omega_{RF} t + 2V_{RF}V_{LO} \cos \omega_{RF} t \cos \omega_{LO} t + V_{LO}^2 \cos^2 \omega_{LO} t] \\ &= \frac{a_2}{2} [V_{RF}^2 (1 + \cos 2\omega_{RF} t) + V_{LO}^2 (1 + \cos 2\omega_{LO} t) \\ &\quad + 2V_{RF}V_{LO} \cos(\omega_{RF} - \omega_{LO}) t + 2V_{RF}V_{LO} \cos(\omega_{RF} + \omega_{LO}) t] \end{aligned}$$

Mixer

Ideal Mixer

up converting

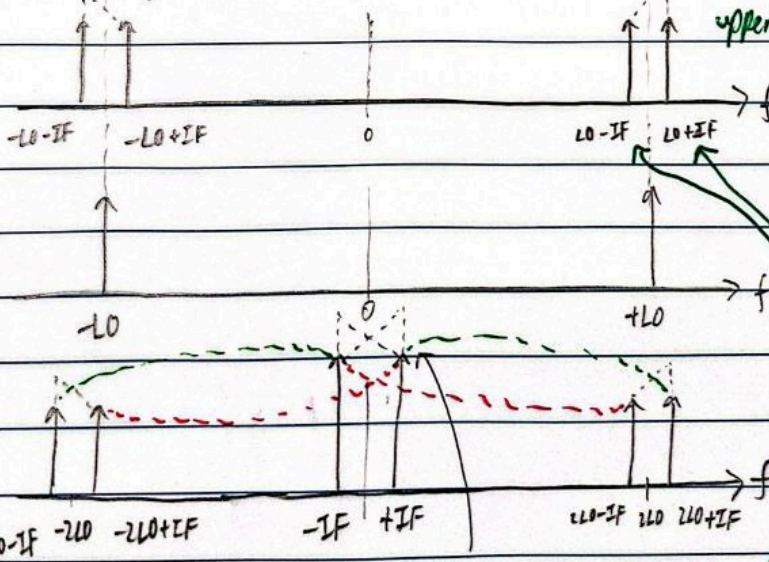


(X)

usually taken as the output

since can use LPF, else can take upper side band & BPF

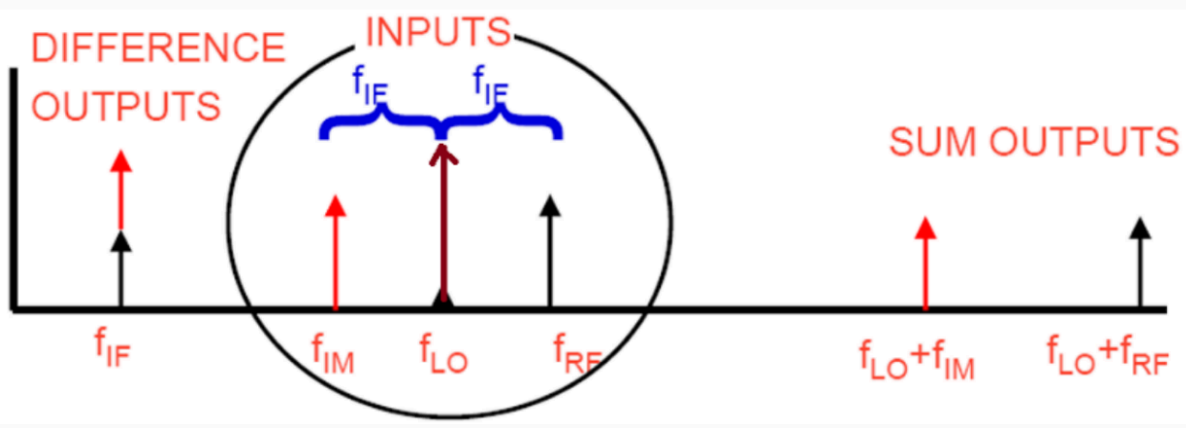
down converting

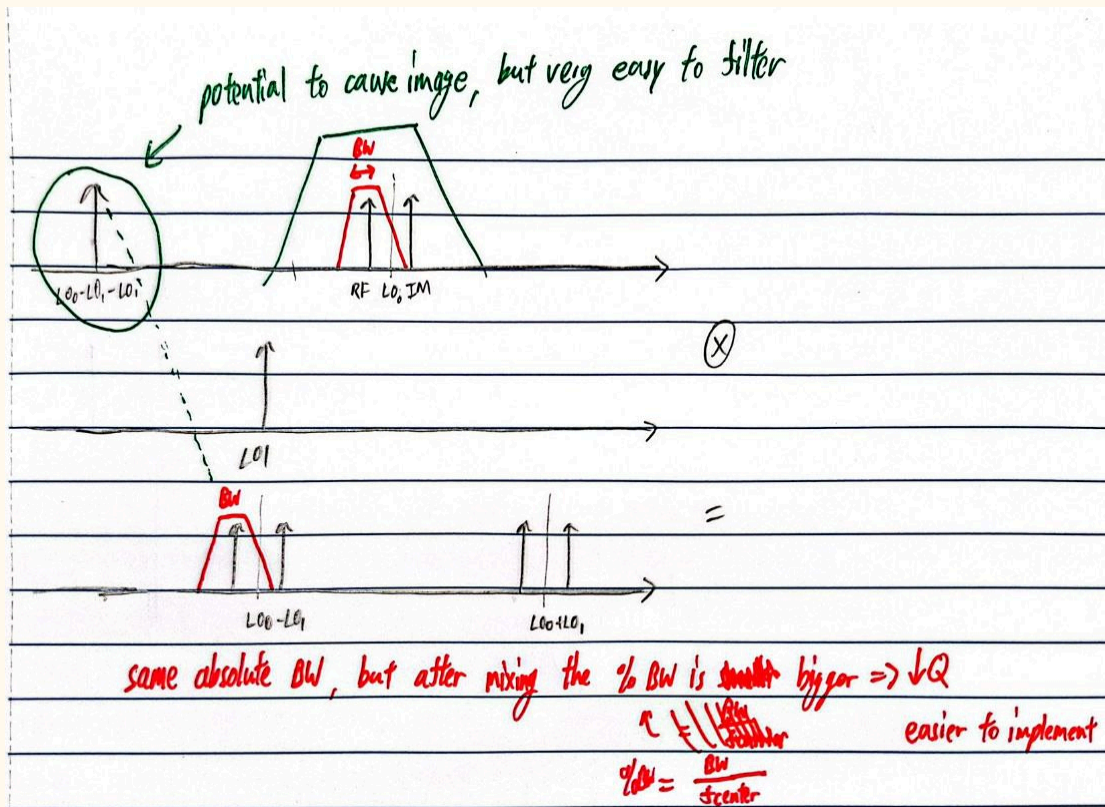


(X)

either side

image overlap $\Rightarrow$ need filter one side first





- downconvert twice can reduce filter Q requirement

- **Conversion loss** of a mixer is defined as the ratio of available input power to the available output power, expressed in dB.

$$L_c = 10 \log \frac{\text{available input power}}{\text{available output power}} \geq 0 \text{ dB}$$

- **Isolation** is a measure of the leakage between ports. LO-IF and LO-RF leakage figures are the most significant. Isolation levels vary from about 25 dB to 70 dB
- **Dynamic range**: Power range over which the mixer operates without serious degradation. Above this range power saturates; below this range noise dominates.

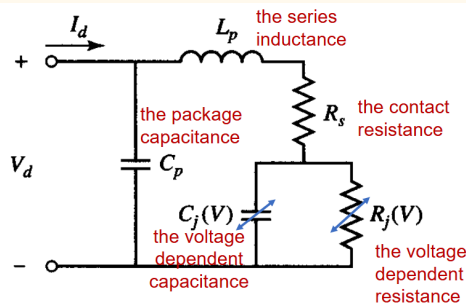
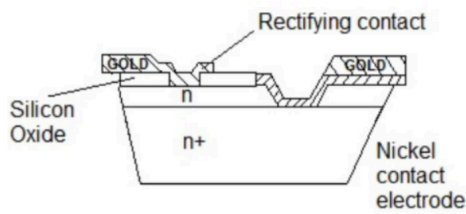
$$\text{Noise floor} = kTB = N_i$$

$$\text{Minimum Detectable Signal (MDS)} = N_i + 3 \text{ dB} \\ = -111 \text{ dBm} + 10 \log(BW \text{ in MHz})$$

$$\text{Dynamic range} = P_D (1 \text{ dB compression point}) - \text{MDS}$$

- compression point depends on which port is being used as the input
- the -111 dBm, is taking T as T0

Microwave Mixer Diode & Equivalent Circuit



Schottky diode (low capacitance) is normally used.
The gate-channel junction of GaAs FET is a Schottky diode junction.

Conversion Loss, L_c

$$L_{c, dB, downconvert} = 10 \log \left(\frac{P_{RF}}{P_{IF}} \right) \quad L_{c, dB, upconvert} = 10 \log \left(\frac{P_{IF}}{P_{RF}} \right)$$

$$L_{c, dB} = L_{d, dB} + L_{m, dB} + L_{h, dB}$$

Diode Loss, L_d

$$L_{d, dB} = 10 \log \left(1 + \frac{R_s}{R_j} + (\omega C_j)^2 R_s R_j \right)$$

Mismatch Loss, L_m

$$L_{d, dB} = -10 \log [(1 - |\Gamma_{RF}|^2)(1 - |\Gamma_{IF}|^2)]$$

$(1 - |\Gamma|^2)$ is the $\frac{\text{transmitted power}}{\text{incident power}}$

Harmonic Loss or Intrinsic Loss, L_h

Due to generation of harmonics, not all the power will be converted into desired signals

SSB versus DSB

SSB versus DSB Conversion Loss:

- DSB signal has both side-bands so it has twice as much power compared to SSB signal.
- Conversion loss is 3 dB less than that of SSB signal

$$(P_{IF})_{DSB} = 2 (P_{IF})_{SSB}$$

$$(L_c)_{DSB} = (L_c)_{SSB} - 3 \text{ (dB)}$$

Assume down-conversion:

$$\begin{aligned} (L_c)_{DSB} &= 10 \log \frac{P_{RF_input}}{P_{IF_output}} \\ &= 10 \log \frac{P_{RF_input}}{2 * (P_{IF_output})_{SSB}} \\ &= 10 \log \frac{P_{RF_input}}{(P_{IF_output})_{SSB}} + 10 \log \frac{1}{2} \\ &= (L_c)_{SSB} - 3 \text{ (dB)} \end{aligned}$$

SSB versus DSB Noise Figure:

- Noise figure for DSB is lower than SSB by 3 dB

$$(F_m)_{DSB} = (F_m)_{SSB} / 2$$

$$(F_m)_{DSB} = (F_m)_{SSB} - 3 \text{ (dB)}$$

DSB →
less loss;
less noise

<https://blog.minicircuits.com/iq-mixers-image-reject-down-conversion-single-sideband-ssb-up-conversion>